

Hydraulic Structures as Flow Measuring Devices

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Abstract – Discharge measurement is the base of proper water management. The effective design and operation of hydraulic structures under both normal and extreme flow conditions depend on the quality of hydrological data. Understanding the water system requires consistent and long-term measurement. Despite that, the gauging station network is sparse, and its numbers are declining worldwide. This article aims to draw attention to the possibility of accurate flow measurement using existing hydraulic structures. Flow over a hydraulic structure profile is a physically well-defined phenomenon as the construction shape is fixed and simple compared to river profiles. The discharge can be derived from rating curves, turbine characteristics, and several easily measured variables. That allows continuous discharge measurement. The accuracy is compared with the gauging station on the river. Suitable technical solutions for ensuring and monitoring ecological flow are discussed.

Keywords – Capacity; differential pressure method; discharge measurement; ecological flow monitoring; rating curve

Nomenclature

μ	Discharge coefficient of the structure	–
ρ	Density of water	kg m ⁻³
B	Effective width of the spillway	m
g	Gravitational acceleration	m s ⁻²
H_{crest}	Overflow crest level	m a. s. l.
H_{gate}	Gate edge level	m a. s. l.
H_{res}	Reservoir water level	m a. s. l.
H_{tail}	Tailwater level	m a. s. l.
Q	Discharge	m ³ s ⁻¹

1. INTRODUCTION

Water management impacts societies environmentally, socially, and financially by influencing crucial issues such as water resource reliability, energy production, or flood hazard prediction [1]. All those issues are gaining importance with advancing climate change. Understanding the water system requires long-term and consistent hydrologic monitoring.

Hydrologic monitoring is based on in situ measurement networks. The lack of data led to a search for an alternative solution, like satellite-based observations [2]. Although the dynamic

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development of remote measurement techniques, the fundamental source of the data is still land-based monitoring [1], [3].

This article aims to draw attention to the possibility of accurate flow measurement using existing hydraulic structures. The purpose of this paper is not to provide full calculation guidelines but rather to provide practicing engineers and owners/operators of large hydraulic structures information about discharge observation. The ability to derive the discharge from values that can be measured easily and continuously is of particular interest.

2. METHODS AND METHODOLOGY

2.1. Discharge Measurement Network

Measurement networks are unevenly distributed worldwide and sparse, which applies primarily to remote areas with difficult access or security concerns [4]. The number of stations peaked in the eighties, followed by a steep decline. Vörösmarty [1] shows that the number of discharge stations declines in various parts of the world, with the most significant drop in Africa (90 % between 1996 and 2001), Russia (25–30 % since 1985), and Post-Soviet countries (loosing from 15 % to 60 %). The decrease is significant also in well-covered countries such as Canada (25 % reduction since 1990). The Global Runoff Data Centre (GRDC) [5] map documents the current situation.

The reduction in the number of gauging stations has various causes (e.g., the collapse of the Soviet Union). One of the significant reasons is that keeping in situ monitoring network is still financially demanding. The data from USGS and Environment Canada shows that operating one station yield a total of \$ 20 000 annually (considering 2001 prices) [3].

This paper draws on the situation in Georgia. As a typical Post-Soviet country, it used to have a well-covering hydrological network that fell apart in the '90s due to political and economic changes. The number of observation sites dropped by almost 80 % [6]. Recently the situation has improved as the NEA (National Environmental Agency) is widening the network of monitoring points. According to the report [7], there were 116 monitoring points on 63 water bodies in 2018.

Despite that, there has been a significant data gap in the last 30-years period. Hannah [8] uses the existing Khertvisi observation point data to show changes in average monthly discharge during last century. Those changes are caused both by global warming that strongly affects the Caucasus region and significant changes in water consumption due to socio-economic changes. Missing relevant data in the last thirty-years period can lead to inappropriate and ineffective hydraulic structure operation.

On the other hand, in Georgia, there are tens of large hydraulic structures with hydropower plants (HPP) in operation. They generate about 90 % of Georgia's electricity production using just about 20 % of the country's hydro energetic potential [9]. This article aims to draw attention to the fact that those structures themselves can serve as an accurate flow-measuring device.

2.2. Discharge Measurement Accuracy

Initially, we need to determine the achievable accuracy of flow measurement. Technical limitations that affect both methods are discussed.

A typical flow measurement device is a gauging station, where the water level is systematically observed. Discharge is then derived by hydrological methods, e.g., the velocity-area method. The flow is usually not measured directly but derived from observable flow conditions – flow heights or width. The U.S. Geological Survey uses four accuracy

classifications to rate station records. The rating ‘excellent’ (1st category) means that about 95 % of daily discharges are within 5 % of the real daily discharges, and ‘good’ (2nd category) are within 10 % of the true daily discharges [10]. These are the rates we want to achieve when measuring discharge on hydraulic structures.

The case study [11] shows an overall measurement error of velocity-area method can range from 6.2 % to 42.8 % at the 95 % confidence level, with an average value of 25.6 %. Those data are relevant for the specific site, and such high error is the exception, not the norm [3]. This case study draws attention to the uncertainties in flow measurement on rivers that must be considered. It presents the leading causes of uncertainties affecting the measurement derived by the velocity-area method:

- Unsteady flow during measurement (especially for high discharges);
- Wind affecting the surface water velocity;
- Inaccuracy of the measuring instrument itself;
- Cross-section measurement errors;
- Spatial variation of the flow;
- Seasonal variation of river bed roughness;
- Imperfect observation of the river profile and changes due to sediment transport and erosion (especially for alluvial rivers).

Accurate discharge monitoring, therefore, relies on the calibration of the discharge rating curve. The calibration must be repeated to include changes in the river profile due to sediment transport, vegetation growth, and ice formation [3].

Flow measurements on hydraulic structures eliminate most of the uncertainties mentioned above. The flow is derived from the level in the reservoir, which changes very slowly and can be measured with high accuracy. The shape of the structures that make up the flow profile is well-defined and constant. Therefore, it is not necessary to repeat the calibration.

On the contrary, the accuracy of the measurement is affected by the following:

- Uncertainties in the shape of structures (difference between project and realization);
- Inaccuracy of the gate position measuring;
- Inaccuracy discharge coefficient assessment.

Those uncertainties and their mitigation will be discussed in the results.

2.3. Discharge Over Hydraulic Structure

Flow over a hydraulic structure profile is a physically well-defined phenomenon as the construction shape is fixed and simple compared to river profiles. Almost any hydraulic structure can serve for flow measurement [12]. High structures where tail-water does not influence the discharge are preferable.

We will deal with calculating the flow over a large dam body. Large dams usually have several discharge objects, as shown in Fig. 1. The Eq. (1) represents the fact that the total outflow from the reservoir is the sum of partial flows through an individual object.

$$Q_{\text{out}} = \sum Q_i \quad (1)$$

Once we estimate the reservoir outflow, we can get to the inflow estimation. Water balance in a reservoir consists of several components – rainfall-runoff, rainfall on the reservoir's water surface, consumption (demand), evaporation, infiltration to the bottom of the reservoir, and discharge [13]. The difference between total inflow and outflow results in changes in reservoir water storage Eq. (2). Curves of flooded areas and volume can be derived with precision which depends on the available map documentation or photogrammetry.

$$\int_{t_1}^{t_2} Q_{in} dt = \Delta V_{accum} + \int_{t_1}^{t_2} Q_{out} dt, \quad (2)$$

where

Q_{in} inflow to reservoir including the rainfall on the water surface;
 Q_{out} outflow from reservoir including the evaporation and infiltration;
 V_{accum} volume of water accumulated in reservoir.

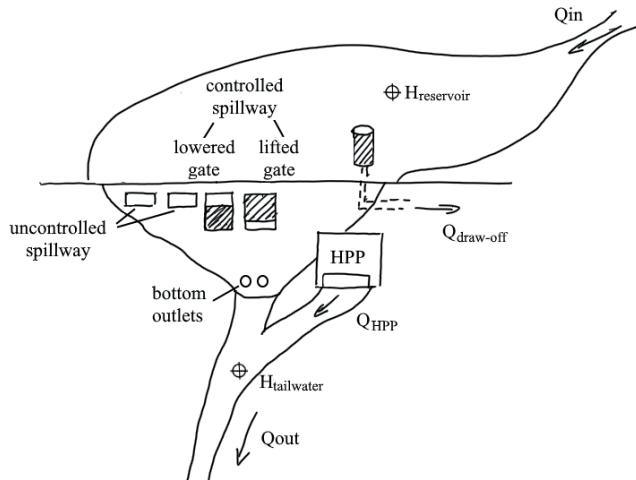


Fig. 1. Dam discharge objects scheme.

2.4. Rating Curve Estimation

The rating curve describes the relationship between the head and discharge. That allows us to derive the discharge directly from the water level measurement. The rating curve depends on the constant construction properties such as type, shape, and surrounding terrain. The discharge coefficient represents those properties and plays a crucial role in defining the rating curve [12].

The discharge coefficient and, subsequently, the rating curve must be determined for each object individually. Analytical formulas describe the derivation of the discharge coefficient in detail. Hydraulic structures books [14], [15] describes the appropriate determination of flow coefficients for different structures in detail. These relationships are accurate for standard construction. It is reasonable to verify the rating curve by a physical or numerical model for atypical configurations.

Laboratory experiments provide an appropriate estimation of the discharge coefficient, but they are expensive [12]. In an extensive review summarizing the modelling of spillways in Sweden, Yang states that a physical model's average price is \$ 50 000–300 000, depending on the range of tests [16]. He also shows excellent agreement between physical and CFD simulations concerning the rating curve when using high-quality grid and proper boundary conditions [17]. The review concludes that CFD modelling is a cheaper way to obtain a rating curve. The application of CFD models for rating curve investigation is cover by various publications [18]–[20]. In contrast, the physical models are still more reliable in identifying complicated and unexpected problems (especially unsteady flow patterns).

2.4.1. Overtaken Structures – Spillways and Gates

Smetana defines discharge over a structure by the Eq. (3) showing that discharge is influenced by spillway (gate) shape represented by the flow coefficient μ and the effective width of the spillway B . Those values are constant or can be expressed as a function of the head [15] (the velocity head must be considered for shallow reservoirs):

$$Q = \frac{2}{3} \mu B \sqrt{2g} (H_{\text{res}} - H_{\text{gate}})^{\frac{3}{2}}, \quad (3)$$

where

μ discharge coefficient;
 B effective width considering the side contraction of the flow;
 H_{res} water level in the reservoir;
 H_{gate} edge of the overspilled structure.

Discharge can be directly derived from one measured variable (H_{res}) for ungated spillways or two measured variables (H_{res} , H_{gate}) for gated spillways.

2.4.2. Underflow Structures – Lifted Gates

From the hydraulic point of view, there is a significant difference between the lowered and lifted gates that behaves as a large opening. The opening is under the water level, and the discharge [15] can be calculated as Eq. (4):

$$Q = \frac{2}{3} \mu B \sqrt{2g} \left((H_{\text{res}} - H_{\text{crest}})^{\frac{3}{2}} - (H_{\text{res}} - H_{\text{gate}})^{\frac{3}{2}} \right), \quad (4)$$

where

μ discharge coefficient;
 B structure width;
 H_{res} water level in the reservoir;
 H_{gate} lower edge of the lifted gate;
 H_{crest} upper edge of the fixed structure.

From Eq. (4), we can see that the water level in the reservoir and the gate position are sufficient for discharge derivation.

2.4.3. Bottom Outlets

The rating the flow coefficient μ defines the rating curve of the lower outlets. The coefficient is provided by the valve producer or can be estimated from the literature. It is reasonable to use the CFD simulation for atypical installation or when higher precision is needed. The flow through the bottom outlets is the function of the head and the position of the bottom outlet gate in Eq. (5).

$$Q = \mu S \sqrt{2gH_{\text{net}}}, \quad (5)$$

where

μ discharge coefficient (function of the valve opening);
 S reference cross-section area (usually pipe cross-sectional area);
 H_{net} net head in front of the valve;

H_{res} water level in the reservoir;
 H_{tail} tail water level.

The hydraulic losses assessment is the most challenging part we are facing when specifying the bottom outlets rating curve. These losses gain importance with the rising velocities in the bottom outlet's channels. We can derive the overall coefficient from a pressure drop in the pipe, which could be measured on-site.

2.5. Hydropower Plant

The hydroelectric power plant is, in fact, a very accurate flow meter. In addition, the operational data of power plants are well logged, even on hydraulic structures with missing monitoring. That eventually enables us to partially reconstruct the historical flow data.

2.5.1. Calculation from the Turbine Efficiency Curve

The turbine flow relates [21] to the measured power output by the head and the overall conversion efficiency from water potential energy to electricity Eq. (6):

$$P_{\text{out}} = \eta_{\text{tot}} H_{\text{net}} \rho g Q, \quad (6)$$

where

P_{out} power output measured on generator terminals;
 η_{tot} total efficiency (turbine, generator, other parts);
 H_{net} net head.

The turbine efficiency curve can be used to derive the turbine flow. The turbine manufacturer provides the efficiency curve, or better, the curve is verified during the guarantee measurement. It describes the relationship between the guide vane and runner blade position, head, and turbine efficiency. These data are unique to each turbine. We can derive the relationship between guide vane opening or power output and the corresponding flow rate for the given net head. An example is presented in Fig. 2. This method assumes that the turbine operates in an optimal runner and guide vanes position.

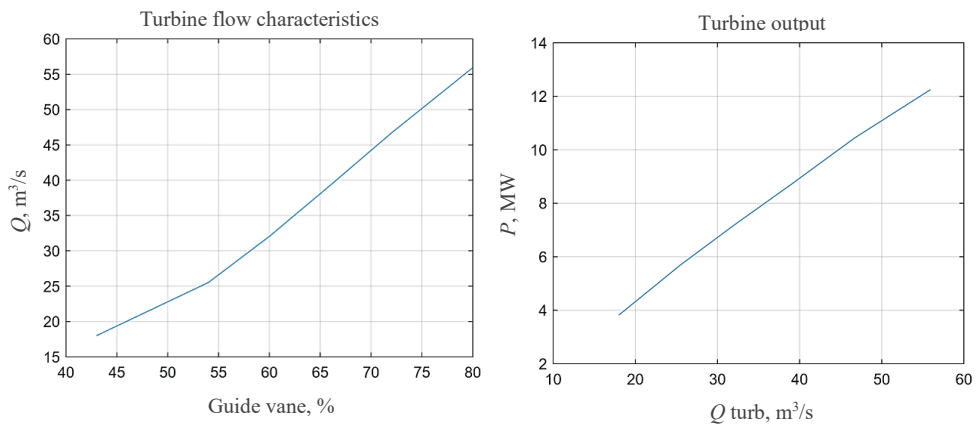


Fig. 2. Mass flow calculation from the turbine efficiency curve for nominal head.

2.5.2. Pressure Difference – Index Method

Pressure difference–index method (also called the Winter-Kennedy method) is a method to derive discharge from the pressure differential between two taps located at a radial section of a spiral casing. The relationship between the pressure difference in a spiral casing and the mass flow is generally given by the equation [22].

$$Q = K_{WK} \Delta p^n, \quad (7)$$

where K_{WK} is the flow coefficient called the Winter-Kennedy constant, and n is the exponent usually kept at 0.5 (can vary between 0.48 and 0.52).

Fig. 3 shows the measurement taps position and graph of mass flow expressed as a function of the pressure difference. This method can be used to precisely derive the relationship between the guide vanes opening and discharge, as shown in Fig. 3.

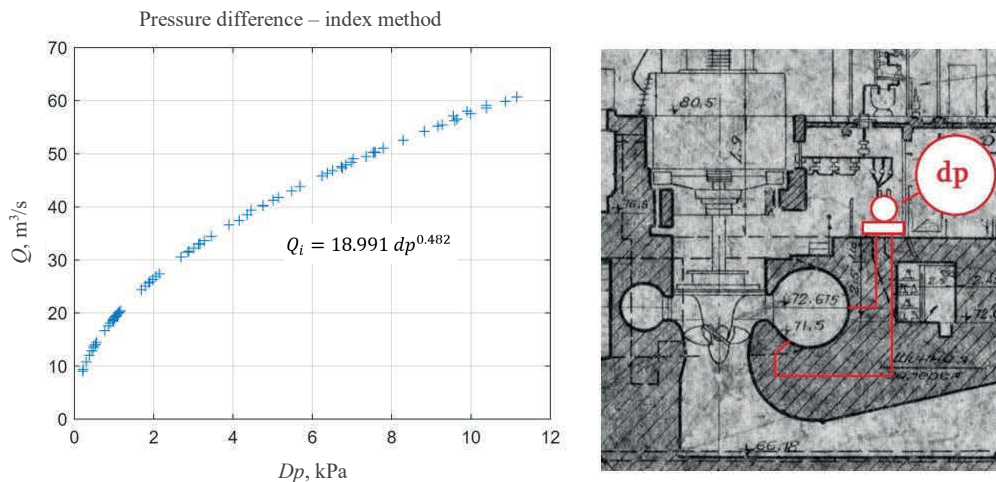


Fig. 3. Winter-Kennedy method scheme – flow measurement.

2.6. Miscellaneous Structures

Dams often secure water supply for industry, irrigation, or drinking water systems. The amount of water abstracted is usually measured depending on the abstraction facility type and the subsequent water transport. Water taken directly from the reservoir should be included in the overall reservoir balance.

3. RESULTS

The summary of the hydro-technical calculation shows that only several variables has to be measured and logged to derive the discharge through the hydraulic structure profile. These include the water level in the reservoir, the tail-water level, and the position of the gates and valves. Relevant variables related to the type of discharge object are given in Table 1. These measurements do not demand high investment or operating costs and allow accurate estimation discharge through the hydraulic structure.

TABLE 1. MONITORING TO DISCHARGE CALCULATION ACCORDING TO ANALYTICAL FORMULAS

Facility	Type	Measured variable
Spillway	Uncontrolled	H_{res}
	Controlled	$H_{\text{res}}, H_{\text{gate}}$
Bottom outlets		$H_{\text{res}}, H_{\text{tail}}, a_{\text{valve}}$
HPP		$H_{\text{res}}, H_{\text{tail}}, P_{\text{out}}, \text{guide vane position}$
Water draw-off		depending on type

Large hydraulic structures mostly have hydropower use and are equipped at least by elementary measurement devices. HPPs already measure and log data (e.g., level in the reservoir, the power delivered to the network). These data are often available also retrospectively and allow a partial reconstruction of flows in previous years.

It is in the interest of all participating subjects to measure the flow through the profile of the hydraulic structure. This data's measurement, recording, and long-term storage are the basis for efficient water management design. In a widely cited article, Vörösmarty [1] concludes from various sources that between 1980–2000 73 % of hydraulic structures in the developing world were over-designed due to unavailable or unreliable hydrologic information.

3.1. Accuracy of Discharge Measurement on Hydraulic Structures

3.1.1. Hydraulic Structures

Physical model studies provide excellent agreement with real structures' behaviour, and they are still considered as a basis for the discharge measurement. They serve as a benchmark for assessing other methods. CFD modelling is experiencing dynamic development and offers significant time and cost savings compared to physical model studies [23].

Numerous works published in professional journals [16], [20], [23], [24] and experience from our applied research show that the rating curve determined by the CFD model corresponds with the values measured on the physical model. The relative percent difference between values from CFD and physical model is below 5 %. The difference between the physical and mathematical models can occasionally increase to ten percent for particularly complicated flow. Even in these cases, the accuracy is comparable to the measurement accuracy in the river, as mentioned in chapter 2.2 (rating 'good' – below 10 % of actual discharge).

It needs to be mentioned that several phenomena can influence the accuracy of the measurement

- Very low overflow height;
- Negative pressure and flow instabilities;
- Rating curve changes due to the material transport (blocking of the openings).

3.1.2. Hydropower Plant

It is the precision of the flow measurement that usually limit the accuracy of determining the turbine efficiency characteristics. A number of methods for measuring turbine flow are described in the literature with different precision [21]. In general, these methods are very accurate. Norms for the pressure-method estimate the measurement error below 2.3 % in efficiency and discharge, respectively [25]. While deriving the discharge backward from the operational data, the measurement errors of the individual input variables enter the

calculation. Although, the total error in flow measurement is at most 5 %. In addition, the operating characteristics of the turbine allow cross-checking between operating values

$$Q_{\text{turb}} = f(P_{\text{gen}}, H, \text{guide vane position}), \quad (8)$$

and at the same time

$$Q_{\text{turb}} = f(n_{11}, Q_{11}, \text{guide vane position}), \quad (9)$$

where

Q_{turb} turbine discharge;
 P_{gen} power output measured on the generator;
 H head;
 n_{11} unit speed;
 Q_{11} unit discharge.

3.2. Ecological Flow Measurement

The ecological flow measurement through the affected part of the river is a particular challenge. It is necessary to take technical measures to ensure continuous environmental flow in the river with relatively small losses in energy production [26]. Research on this topic usually focuses more on determining the required flow rate [26]–[28]. We also need to describe the reliable technical measures that ensure the environmental flow as they are crucial for everyday operation.

One of the technical solutions that secure the environmental flow and reduce losses of produced energy is the appropriate choice of flow conversion. Fig. 4 shows the situation where the ecological flow is passing the dam spillways. We expect the fluctuation in the reservoir ± 1 m due to the inflow variation and HPP operation. At the same time, we assume that the position of the gates is not being adjusted, which is likely wherever an automatic system does not control the gate position.

The flow over lowered gate shows considerable flow changes due to the water level fluctuation. This technical solution does not reliably maintain sufficient flow in the affected section of the river.

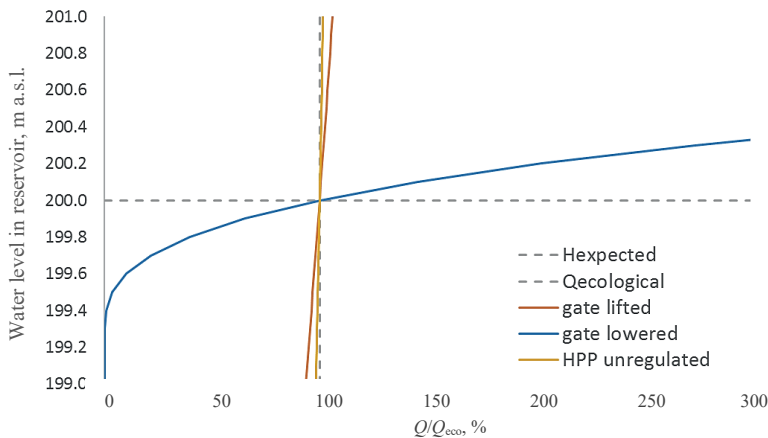


Fig. 4. Comparison of ecological flow discharge facilities – flow difference due to the reservoir water level fluctuation.

When the water level drops, the flow in the river could even drop to zero. At the same time, the situation is uncomfortable either for the HPP operator. The rise in water level leads to loss of a large volume of water that flows over the spillway and is not utilized by the HPP. This situation increases the risk that the operator will permanently limit the flow to the river section to reduce its losses.

Conversely, using lifting gates, bottom outlets or HPP, the flow is almost constant regardless of the water level in the reservoir. Therefore, leaving the gates in one position for long periods is possible. It is convenient for the operator, but at the same time, it rapidly increases ecological flow discharge reliability.

3.3. Hydropower Plant as an Ecological Flow Discharge Structure

An ideal solution is energy utilization of ecological flow by small run-off hydroelectric power plant. It makes sense especially on large dams that do not allow any fish-pass. The transfer of water by HPP ensures a very stable ecological flow. Even the simple unregulated turbine has minimal flow fluctuation due to the reservoir water level changes. A comparison of flow fluctuations for different discharge structures is shown in Fig. 4. The financial profit motivates the hydraulic structure operator not to limit the ecological flow. The production of green energy is then a positive side benefit.

4. CONCLUSION

Flow measurement is the basis for understanding the water system and ensuring the effective design and operation of hydraulic structures. We propose to use the existing hydraulic structures for additional discharge measurement devices besides the classic gauging stations.

Structures' rating curves enable us to derive the discharge from several easily measurable values, such as the water level in the reservoir or gate position. The rating curve determined from table values can be calibrated using physical and mathematical modelling. Structure shape and hydraulic properties are stable and well-defined, and the rating curve does not need periodic re-calibration. Even extreme flow rates can be measured accurately and safely with this system.

This approach can also be applied to ecological flow monitoring. Additionally, the small run-off HPP is the ideal solution for ecological flow discharge on sites where the fish pass is not applicable. On sites where the small HPP is not feasible, technical measures must ensure that the reservoir water level does not strongly influence the ecological flow. Lifted gates or bottom outlets secure the environmental flow while reducing the water losses for energy production.

The initial investment in rating curves' creation and calibration brings a continuous overview of the operation of the hydraulic structure and reservoir. All interested sides will benefit from the improved water management – the local water ecosystems by keeping the sufficient residual flow in the rivers and increasing water quality, energy producers by reducing water losses, and the hydraulic structures operators by increasing safety during extreme flood events.

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